ML-BASED INTRUSION DETECTION FOR HEALTHCARE NETWORKS

University of Cincinnati
EECE Senior Capstone | Spring 2025

01 | PROJECT OVERVIEW

THE PROBLEM

Healthcare networks are the most targeted sector in cybersecurity — accounting for 25% of all ransomware attacks. Traditional signature-based IDS solutions cannot detect novel, zero-day threats without constant manual updates.

Connected medical devices (IoT) running legacy firmware create persistent attack surfaces that put patient data and critical care systems at risk.

OBJECTIVE

Build and demonstrate a machine learning-powered IDS that detects network intrusions in real time. Using virtual machines simulating a healthcare network, we compare traffic on an unprotected network against one defended by our Suricata + ML detection pipeline — showing the security gap live.

25%	$10.9M	RT
of all ransomware attacks target healthcare	average breach cost in the healthcare sector	real-time threat detection with ML classification

TECHNOLOGIES USED

Suricata	Real-time IDS & network traffic capture
scikit-learn	ML model training & classification
Kali Linux	Attack simulation platform (attacker VM)
Wireshark	Packet analysis & traffic verification
Python	Data pipeline, preprocessing & inference
VirtualBox	Virtualized demo environment

WHY HEALTHCARE?

- Ransomware attacks have halted hospital operations, rerouted ambulances, and delayed surgeries.
- Medical IoT devices rarely receive security patches — creating permanent, exploitable vulnerabilities.
- HIPAA mandates breach notification — a single incident costs millions in fines and remediation alone.
- An adaptive ML-based IDS detects behavioral anomalies without requiring constant signature updates.

02 | DATA PIPELINE

01 NETWORK TRAFFIC CAPTURE

Suricata captures live packet flows in the simulated healthcare VM network. PCAP files record both normal operations and simulated attack scenarios to build a labeled training dataset.

02 FEATURE EXTRACTION

Flow-based features are computed per connection: packet count, byte count, flow duration, inter-arrival times, TCP flag distribution, protocol type, and destination port number.

03 PREPROCESSING

Raw features are cleaned, normalized with MinMaxScaler, and label-encoded. Class imbalance is addressed via SMOTE. Dataset is split 80% training / 20% testing with stratified sampling.

04 MODEL TRAINING

Random Forest classifier is trained on labeled feature vectors. GridSearchCV tunes hyperparameters (n_estimators, max_depth, min_samples_split). Evaluated on precision, recall, F1-score, and ROC-AUC.

05 REAL-TIME INFERENCE

The trained model is serialized with joblib and integrated into Suricata's EVE JSON alert pipeline. New network flows are classified within milliseconds of capture using a Python listener.

06 ALERT & RESPONSE

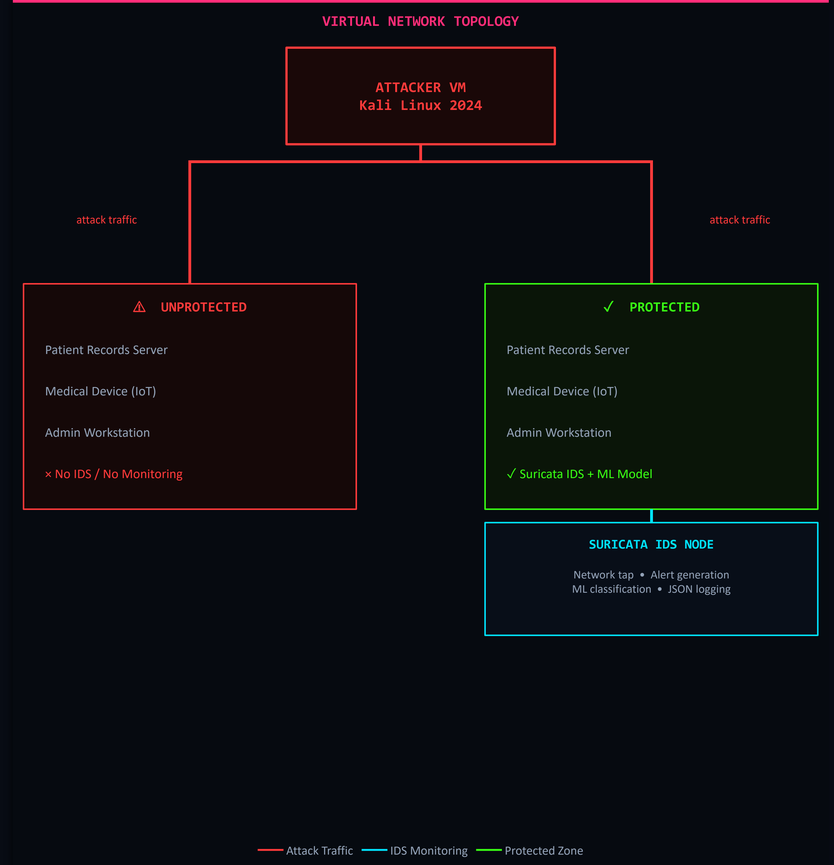
Detected intrusions trigger categorized alerts (Port Scan, DoS, Brute Force, ARP Spoof). Alerts are logged in structured JSON, displayed on the demo dashboard, and compared against the unprotected network.

03 | NETWORK ARCHITECTURE

OVERVIEW

A virtual network environment with three segments: Attacker (Kali Linux), Unprotected Target, and Protected Target (Suricata IDS). Both target networks receive identical attack traffic for direct side-by-side comparison.

VIRTUAL NETWORK TOPOLOGY



VM CONFIGURATION

Attacker VM:	Kali Linux 2024.1 — NAT network adapter
Target VMs:	Ubuntu Server 22.04 — internal network
IDS Node:	Suricata 7.x on dedicated Suricata laptop
Hypervisor:	VirtualBox — Windows host
Network Mode:	Host-only adapter with packet tap
Monitoring:	Wireshark + Suricata EVE JSON logs